

STAT8289: Assignment 3

October 2026

Overview

This report analyzes survey data from 877 customers concerning their preferred long-distance telephone provider and their satisfaction with three provider attributes:

- Y : preferred provider, with levels A, B, and C;
- X_1 : satisfaction with the provider's reputation as an industry leader;
- X_2 : satisfaction with price;
- X_3 : satisfaction with variety of services.

The three satisfaction variables are coded as **low** and **high**. All four variables were converted to factors because **bnlearn** treats discrete Bayesian-network nodes as categorical variables and uses their factor levels to define the states of the corresponding multinomial distributions.

Unless stated otherwise, the global precision parameter is

$$\alpha = 12.$$

1 Part A: Guide to the bnlearn Functions

1.1 DAGs and Factorization

A Bayesian network represents the joint distribution by a directed acyclic graph (DAG). If Π_i denotes the parent set of node Y_i , the joint distribution factorizes as

$$p(y_1, \dots, y_I) = \prod_{i=1}^I p(y_i \mid \Pi_i).$$

In a **bnlearn** model string, a node with no parents is written as

[Y],

while a node with parents is written as, for example,

$$[X_2|Y:X_1].$$

This means that X_2 has parents Y and X_1 , corresponding to the factor

$$p(x_2 | y, x_1).$$

The five candidate model strings used in Part B are

$$\begin{aligned} M_1 : & \quad [Y] [X_1|Y] [X_2|Y:X_1] [X_3|Y:X_1:X_2], \\ M_2 : & \quad [Y] [X_1|Y] [X_2|Y] [X_3|Y], \\ M_3 : & \quad [Y] [X_1|Y] [X_2|Y:X_1] [X_3|Y], \\ M_4 : & \quad [Y] [X_1|Y] [X_2|Y] [X_3|Y:X_1], \\ M_5 : & \quad [Y] [X_1|Y] [X_2|Y] [X_3|Y:X_2]. \end{aligned}$$

`model2network()`

`model2network()` converts a model string into a DAG object.

Example:

```
M3 <- model2network("[Y] [X1|Y] [X2|Y:X1] [X3|Y]")
```

The resulting graph represents

$$p(y, x_1, x_2, x_3) = p(y)p(x_1 | y)p(x_2 | y, x_1)p(x_3 | y).$$

`modelstring()`

`modelstring()` performs the reverse operation and returns the model-string representation of a DAG.

Example:

```
modelstring(M3)
```

The order of parent names in the printed string may differ from the input string, but the DAG and therefore the factorization are unchanged.

`naive.bayes()`

`naive.bayes()` constructs a naive Bayes classifier. The `training` argument specifies the class node.

Example:

```
nb <- naive.bayes(dat, training = "Y")
```

The resulting model is

$$[Y] [X1|Y] [X2|Y] [X3|Y],$$

which is exactly M_2 . Thus, `naive.bayes()` automatically constructs the DAG in which Y is the parent of every predictor and the predictors are mutually conditionally independent given Y .

`nparams()`

`nparams()` reports the number of free parameters in a Bayesian network.

For the naive Bayes model,

$$(3 - 1) + 3\{3(2 - 1)\} = 2 + 9 = 11$$

free parameters are required.

Example:

```
nparams(nb, data = dat)
```

1.2 Parameter Learning

For a discrete node Y_i , let

$$N_{ijk}$$

denote the number of observations for which node i is in category k and its parents are in configuration j . Also let

$$N_{ij} = \sum_k N_{ijk}.$$

```
bn.fit(..., method="mle")
```

With maximum-likelihood estimation,

$$\hat{\theta}_{ijk}^{MLE} = \frac{N_{ijk}}{N_{ij}}.$$

For example, for $X_1 \mid Y = A$,

$$N_{\text{low},A} = 203, \quad N_{\text{high},A} = 130, \quad N_A = 333.$$

Thus,

$$\hat{P}(X_1 = \text{low} \mid Y = A) = \frac{203}{333} = 0.6096$$

and

$$\hat{P}(X_1 = \text{high} \mid Y = A) = \frac{130}{333} = 0.3904.$$

The corresponding R code is

```
fit_mle <- bn.fit(nb, data = dat, method = "mle")
```

Therefore `method="mle"` implements the estimator N_{ijk}/N_{ij} .

```
bn.fit(..., method="bayes", iss=alpha)
```

For Bayesian parameter learning, `iss` is the imaginary sample size and corresponds to the global precision α .

Under the initial-ignorance prior,

$$\alpha_{ijk} = \frac{\alpha}{C_i Q_i},$$

where C_i is the number of states of node i and Q_i is the number of parent configurations.

The posterior mean is

$$E(\theta_{ijk} \mid D) = \frac{\alpha_{ijk} + N_{ijk}}{\alpha_{ij} + N_{ij}}.$$

For example, for X_1 ,

$$C_1 = 2, \quad Q_1 = 3,$$

so when $\alpha = 12$,

$$\alpha_{ijk} = \frac{12}{2(3)} = 2.$$

For $Y = A$,

$$P(X_1 = \text{low} \mid Y = A, D) = \frac{203 + 2}{333 + 4} = \frac{205}{337} = 0.6083.$$

This is exactly the value returned by

```
fit_nb <- bn.fit(nb, data = dat,  
                method = "bayes", iss = 12)
```

```
coef()
```

For a fitted discrete Bayesian-network node, `coef()` can be used to extract the estimated conditional probability table.

Example:

```
coef(fit_nb$X1)
```

The entries correspond to the posterior mean estimates of the θ_{ijk} parameters.

1.3 Classification Probabilities

For an observed attribute profile $x = (x_1, x_2, x_3)$, classification requires

$$p(Y \mid x_1, x_2, x_3, D).$$

For naive Bayes,

$$p(Y = j \mid x, D) \propto p(Y = j \mid D) \prod_{\ell=1}^3 p(X_{\ell} = x_{\ell} \mid Y = j, D).$$

```
predict(..., prob=TRUE)
```

The option `prob=TRUE` returns the posterior probabilities of the possible classes in addition to the predicted class.

Example:

```
predict(fit_nb, node = "Y",  
        data = new_customer,  
        prob = TRUE,  
        method = "exact")
```

The three relevant inference methods are:

parents: Uses only the parents of the target node. Because Y has no parents in all five candidate DAGs, this method ignores X_1, X_2, X_3 and returns the same marginal probabilities $p(Y \mid D)$ for every customer. It is therefore inappropriate for classification in this assignment.

bayes-lw: Uses likelihood-weighted Monte Carlo simulation. It incorporates the observed predictor values, but the answer changes slightly across runs because of Monte Carlo error.

exact: Performs exact probabilistic inference. It has no Monte Carlo approximation error and is therefore used for the final comparisons.

`cpquery()`

`cpquery()` estimates conditional probabilities such as

$$P(Y = A \mid X_1 = \text{high}, X_2 = \text{low}, X_3 = \text{high})$$

by simulation.

Example:

```
cpquery(fit_nb,
        event = (Y == "A"),
        evidence = (X1 == "high" &
                    X2 == "low" &
                    X3 == "high"),
        n = 100000)
```

Because it is simulation-based, its output changes slightly from one run to another unless a random seed is fixed. Increasing the number of simulations reduces Monte Carlo error.

1.4 Marginal Likelihood

`score()`

The Bayesian Dirichlet equivalent score

```
score(M, data = dat, type = "bde", iss = 12)
```

equals the log marginal likelihood

$$\log p(D \mid M).$$

For a Bayesian network,

$$p(D \mid M) = \prod_i p(D_i \mid M),$$

so

$$\log p(D \mid M) = \sum_i \log p(D_i \mid M).$$

This explains why

```
score(M, data = dat,
      type = "bde",
      iss = 12,
      by.node = TRUE)
```

returns a contribution for each node whose sum equals the total BDe score.

The option

```
type = "bic"
```

computes the BIC score instead. BIC is an asymptotic model-selection criterion and is not the exact log marginal likelihood used for the Bayesian posterior model probabilities in this assignment.

2 Part B: Analysis of the Carrier Data

2.1 B1. Exploration

The four-way contingency table is shown below.

Table 1: Four-way contingency table

Y	X_1	X_2	X_3	Count
A	low	low	low	113
B	low	low	low	98
C	low	low	low	73
A	high	low	low	21
B	high	low	low	60
C	high	low	low	5
A	low	high	low	19
B	low	high	low	8
C	low	high	low	7
A	high	high	low	8
B	high	high	low	8
C	high	high	low	6
A	low	low	high	35
B	low	low	high	18
C	low	low	high	17
A	high	low	high	27
B	high	low	high	66
C	high	low	high	9
A	low	high	high	36
B	low	high	high	9
C	low	high	high	15
A	high	high	high	74
B	high	high	high	113
C	high	high	high	32

The cross-tabulations are

Table 2: Preferred provider versus reputation satisfaction

	$X_1 = \text{low}$	$X_1 = \text{high}$
A	203	130
B	133	247
C	112	52

Table 3: Preferred provider versus price satisfaction

	$X_2 = \text{low}$	$X_2 = \text{high}$
A	196	137
B	242	138
C	104	60

Table 4: Preferred provider versus service-variety satisfaction

	$X_3 = \text{low}$	$X_3 = \text{high}$
A	161	172
B	174	206
C	91	73

The within-provider proportions of high satisfaction are

Table 5: Proportion reporting high satisfaction

	A	B	C
X_1 : Reputation	0.3904	0.6500	0.3171
X_2 : Price	0.4114	0.3632	0.3659
X_3 : Services	0.5165	0.5421	0.4451

Reputation satisfaction, X_1 , appears most strongly associated with provider preference. The high-reputation percentage is 65.0% for B, compared with 39.0% for A and 31.7% for C. In contrast, the price and service-variety proportions are substantially more similar across the three providers.

2.2 B2. Naive Bayes

2.2.1 DAG and Factorization

The naive Bayes DAG is

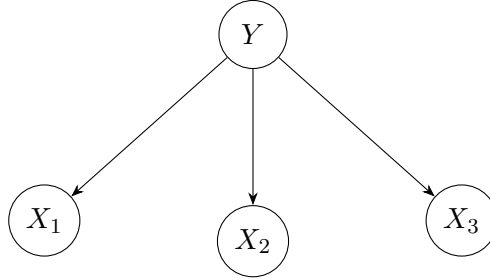


Figure 1: Naive Bayes classifier

Its factorization is

$$p(y, x_1, x_2, x_3) = p(y)p(x_1 | y)p(x_2 | y)p(x_3 | y).$$

Let

$$Y_1 = Y, \quad Y_2 = X_1, \quad Y_3 = X_2, \quad Y_4 = X_3.$$

Then

$$\boldsymbol{\theta}_1 = (P(Y = A), P(Y = B), P(Y = C)),$$

and for $j \in \{A, B, C\}$,

$$\boldsymbol{\theta}_{2j} = (P(X_1 = low \mid Y = j), P(X_1 = high \mid Y = j)),$$

$$\boldsymbol{\theta}_{3j} = (P(X_2 = low \mid Y = j), P(X_2 = high \mid Y = j)),$$

$$\boldsymbol{\theta}_{4j} = (P(X_3 = low \mid Y = j), P(X_3 = high \mid Y = j)).$$

2.2.2 Dirichlet Priors and Posteriors

For Y ,

$$C_Y = 3, \quad Q_Y = 1,$$

so

$$\boldsymbol{\theta}_1 \sim Dir(4, 4, 4).$$

Since the observed counts are

$$(333, 380, 164),$$

the posterior is

$$\boxed{\boldsymbol{\theta}_1 \mid D \sim Dir(337, 384, 168)}.$$

For each binary predictor in the naive Bayes model,

$$C_i = 2, \quad Q_i = 3,$$

so each parent configuration has prior

$$Dir(2, 2).$$

The posterior Dirichlet distributions are

Table 6: Posterior Dirichlet distributions under naive Bayes

Node	$Y = A$	$Y = B$	$Y = C$
X_1	$Dir(205, 132)$	$Dir(135, 249)$	$Dir(114, 54)$
X_2	$Dir(198, 139)$	$Dir(244, 140)$	$Dir(106, 62)$
X_3	$Dir(163, 174)$	$Dir(176, 208)$	$Dir(93, 75)$

The posterior mean CPTs are

Table 7: Posterior mean CPT for $X_1 | Y$

X_1	A	B	C
low	0.6083	0.3516	0.6786
high	0.3917	0.6484	0.3214

Table 8: Posterior mean CPT for $X_2 | Y$

X_2	A	B	C
low	0.5875	0.6354	0.6310
high	0.4125	0.3646	0.3690

Table 9: Posterior mean CPT for $X_3 | Y$

X_3	A	B	C
low	0.4837	0.4583	0.5536
high	0.5163	0.5417	0.4464

2.2.3 Classification by Hand

Consider a customer with

$$X_1 = high, \quad X_2 = low, \quad X_3 = high.$$

For provider A,

$$\begin{aligned}
 s_A &= P(A | D)P(X_1 = high | A, D)P(X_2 = low | A, D)P(X_3 = high | A, D) \\
 &= (0.3790776)(0.3916914)(0.5875371)(0.5163205) \\
 &= 0.045043.
 \end{aligned}$$

For B,

$$s_B = (0.4319460)(0.6484375)(0.6354167)(0.5416667) \\ = 0.096403.$$

For C,

$$s_C = (0.1889764)(0.3214286)(0.6309524)(0.4464286) \\ = 0.017110.$$

Normalizing gives

$$P(A \mid x, D) = \frac{s_A}{s_A + s_B + s_C} = 0.2841,$$

$$P(B \mid x, D) = 0.6080,$$

and

$$P(C \mid x, D) = 0.1079.$$

Therefore,

$$\boxed{\hat{Y} = B}.$$

Exact inference with `predict()` gives exactly the same probabilities.

Using `cpquery()` with 100,000 simulation draws gave approximately

$$(0.2841, 0.6029, 0.1130).$$

The difference is due to Monte Carlo variability.

2.3 B3. Prior Sensitivity

The naive Bayes model was refitted for four values of the global precision parameter.

Table 10: Classification probabilities for $(X_1, X_2, X_3) = (high, low, high)$

α	$P(A \mid x, D)$	$P(B \mid x, D)$	$P(C \mid x, D)$
1	0.2834	0.6111	0.1055
12	0.2841	0.6080	0.1079
100	0.2888	0.5855	0.1257
1000	0.3118	0.4704	0.2179

The posterior mean has the form

$$E(\theta_{ijk} \mid D) = \frac{\alpha_{ijk} + N_{ijk}}{\alpha_{ij} + N_{ij}},$$

which can also be interpreted as a weighted average of the prior mean and the observed sample proportion.

As α increases, the prior receives more weight. Because the initial-ignorance prior is centered at equal probabilities, the posterior CPTs are increasingly shrunk toward uniformity. Consequently, the posterior probability of B decreases from about 0.61 to 0.47, while those of A and C increase. B remains the predicted class for all four values of α , but the classification becomes much less certain when α is very large.

2.4 B4. Five Candidate Classifiers

2.4.1 B4.1 Factorizations and Parent Configurations

For M_1 ,

$$p(y, x_1, x_2, x_3) = p(y)p(x_1 \mid y)p(x_2 \mid y, x_1)p(x_3 \mid y, x_1, x_2).$$

The parent sets are

$$Pa(Y) = \emptyset, \quad Pa(X_1) = \{Y\}, \quad Pa(X_2) = \{Y, X_1\}, \quad Pa(X_3) = \{Y, X_1, X_2\}.$$

For M_2 ,

$$p(y, x_1, x_2, x_3) = p(y)p(x_1 \mid y)p(x_2 \mid y)p(x_3 \mid y),$$

with

$$Pa(Y) = \emptyset, \quad Pa(X_1) = Pa(X_2) = Pa(X_3) = \{Y\}.$$

For M_3 ,

$$p(y, x_1, x_2, x_3) = p(y)p(x_1 \mid y)p(x_2 \mid y, x_1)p(x_3 \mid y),$$

with

$$Pa(X_2) = \{Y, X_1\}.$$

For M_4 ,

$$p(y, x_1, x_2, x_3) = p(y)p(x_1 \mid y)p(x_2 \mid y)p(x_3 \mid y, x_1),$$

with

$$Pa(X_3) = \{Y, X_1\}.$$

For M_5 ,

$$p(y, x_1, x_2, x_3) = p(y)p(x_1 | y)p(x_2 | y)p(x_3 | y, x_2),$$

with

$$Pa(X_3) = \{Y, X_2\}.$$

2.4.2 B4.2 Bayesian Estimation: Example Using M_3

For M_3 , X_2 has parents (Y, X_1) . Hence

$$C_{X_2} = 2, \quad Q_{X_2} = 3 \times 2 = 6.$$

Therefore,

$$\alpha_{ijk} = \frac{12}{2(6)} = 1,$$

so each conditional distribution has prior

$$Dir(1, 1).$$

The posterior distributions are

$$(Y = A, X_1 = low) : Dir(149, 56),$$

$$(Y = A, X_1 = high) : Dir(49, 83),$$

$$(Y = B, X_1 = low) : Dir(117, 18),$$

$$(Y = B, X_1 = high) : Dir(127, 122),$$

$$(Y = C, X_1 = low) : Dir(91, 23),$$

and

$$(Y = C, X_1 = \text{high}) : \text{Dir}(15, 39).$$

The corresponding posterior mean probabilities are

Table 11: M_3 : posterior mean distribution of $X_2 \mid Y, X_1$

Y	X_1	$P(X_2 = \text{low})$	$P(X_2 = \text{high})$
A	low	0.7268	0.2732
A	high	0.3712	0.6288
B	low	0.8667	0.1333
B	high	0.5100	0.4900
C	low	0.7982	0.2018
C	high	0.2778	0.7222

For example,

$$P(X_2 = \text{high} \mid Y = A, X_1 = \text{low}, D) = 0.2732,$$

whereas

$$P(X_2 = \text{high} \mid Y = A, X_1 = \text{high}, D) = 0.6288.$$

Thus, even after conditioning on Y , the distribution of X_2 changes substantially with X_1 . This is evidence against the naive Bayes conditional-independence assumption.

2.4.3 Comparison of Prediction Methods

Using M_3 , the default `parents` method returns the same probabilities for every profile:

$$P(A) = 0.3791, \quad P(B) = 0.4319, \quad P(C) = 0.1890.$$

It therefore predicts B for every profile. This occurs because Y has no parents, so `parents` does not use the observed values of X_1, X_2, X_3 .

Both `bayes-lw` and `exact` use the observed attributes. For M_3 , both methods gave the hard classifications

$$A, B, A, B, A, B, A, B.$$

However, `bayes-lw` gave slightly different posterior probabilities because it relies on likelihood-weighted Monte Carlo simulation. Exact inference was therefore used for the remainder of the analysis.

2.4.4 Posterior Classification Probabilities

Table 12: Posterior class probabilities for all eight profiles

X_1	X_2	X_3	M1-A	M1-B	M1-C	M2-A	M2-B	M2-C	M3-A	M3-B	M3-C	M4-A	M4-B	M4-C	M5-A	M5-B	M5-C
low	low	low	.3975	.3450	.2574	.4240	.2862	.2898	.4093	.3046	.2861	.3962	.3447	.2591	.4285	.2917	.2797
high	low	low	.2457	.6914	.0629	.2910	.5626	.1463	.2627	.6452	.0920	.2553	.6350	.1097	.2915	.5685	.1400
low	high	low	.5493	.2394	.2113	.4715	.2601	.2685	.5635	.1716	.2649	.4433	.3152	.2415	.5239	.1838	.2922
high	high	low	.3617	.3617	.2766	.3335	.5269	.1397	.3412	.4753	.1835	.2949	.5995	.1055	.4140	.4161	.1699
low	low	high	.4965	.2587	.2448	.4418	.3301	.2281	.4252	.3503	.2245	.5228	.2199	.2573	.4429	.3454	.2117
high	low	high	.2657	.6425	.0918	.2841	.6080	.1079	.2510	.6825	.0664	.2984	.5696	.1320	.2789	.6230	.0981
low	high	high	.5935	.1545	.2520	.4900	.2992	.2108	.5909	.1993	.2098	.5702	.1960	.2338	.4711	.3017	.2272
high	high	high	.3379	.5147	.1474	.3262	.5706	.1032	.3392	.5230	.1378	.3415	.5327	.1258	.3135	.5752	.1113

Note: add the following package to the preamble for the landscape table:

`\usepackage{pdfscape}`

The corresponding hard classifications are

Table 13: Predicted classes for all eight profiles

Profile	X_1	X_2	X_3	M1	M2	M3	M4	M5
1	low	low	low	A	A	A	A	A
2	high	low	low	B	B	B	B	B
3	low	high	low	A	A	A	A	A
4	high	high	low	A*	B	B	B	B
5	low	low	high	A	A	A	A	A
6	high	low	high	B	B	B	B	B
7	low	high	high	A	A	A	A	A
8	high	high	high	B	B	B	B	B

The models agree on seven of the eight profiles. The only disagreement occurs for

$$(X_1, X_2, X_3) = (high, high, low).$$

Under M_1 ,

$$P(A \mid x, D) = P(B \mid x, D) = 0.3617021,$$

while

$$P(C \mid x, D) = 0.2765957.$$

Thus A and B are exactly tied. The reported A label is therefore a tie-breaking result rather than evidence that A is more probable than B.

M_5 is also nearly tied for this profile:

$$P(A \mid x, D) = 0.4140, \quad P(B \mid x, D) = 0.4161,$$

a difference of only

$$0.0021.$$

The smallest top-two probability gaps for the five models were

Table 14: Most ambiguous profile under each model

Model	Smallest gap	Profile
M1	0.0000	<i>(high, high, low)</i>
M2	0.1117	<i>(low, low, high)</i>
M3	0.0749	<i>(low, low, high)</i>
M4	0.0515	<i>(low, low, low)</i>
M5	0.0021	<i>(high, high, low)</i>

2.4.5 Confusion Matrices and Accuracy

The confusion matrix for M_1 was

	$\hat{A}$	$\hat{B}$	$\hat{C}$
A	205	128	0
B	139	241	0
C	117	47	0

and the confusion matrix for M_2 – M_5 was

	$\hat{A}$	$\hat{B}$	$\hat{C}$
A	203	130	0
B	133	247	0
C	112	52	0.

The accuracies are

Table 15: In-sample classification accuracy

Model	Accuracy
M1	0.5086
M2	0.5131
M3	0.5131
M4	0.5131
M5	0.5131

None of the five classifiers predicts C as the hard class. This does not mean that the posterior probability of C is zero. Rather, C never has the largest posterior probability among the three providers for any of the eight possible attribute profiles.

Because M_1 has an exact A/B tie for one profile, its hard classification for those cases depends on how ties are resolved. The posterior probabilities themselves remain well defined.

2.4.6 B4.3 Log Marginal Likelihood

The BDe log marginal likelihoods at $\alpha = 12$ are

Table 16: Bayesian model comparison

Model	$\log p(D \mid M)$	$P(M \mid D)$
M1	-2488.666	≈ 1
M2	-2699.460	2.8390×10^{-92}
M3	-2636.569	5.8444×10^{-65}
M4	-2608.371	1.0301×10^{-52}
M5	-2587.904	7.9727×10^{-44}

Because all five models have equal prior probability,

$$P(M_g) = \frac{1}{5},$$

we have

$$P(M_g \mid D) \propto p(D \mid M_g).$$

Therefore,

$$\boxed{P(M_1 \mid D) \approx 1}.$$

The data provide overwhelming evidence in favor of M_1 . In particular,

$$\log p(D \mid M_1) - \log p(D \mid M_5) = 99.238,$$

even though M_5 is the second-best model.

This strongly suggests that substantial dependence remains among X_1, X_2, X_3 even after conditioning on provider preference Y .

2.4.7 Verification of One Node Contribution

For Y , the prior is

$$(\theta_A, \theta_B, \theta_C) \sim \text{Dir}(4, 4, 4)$$

and the observed counts are

$$(333, 380, 164).$$

The node contribution is therefore

$$\begin{aligned} \log p(D_Y \mid M) &= \log \Gamma(12) - \log \Gamma(889) \\ &\quad + \log \Gamma(337) - \log \Gamma(4) \\ &\quad + \log \Gamma(384) - \log \Gamma(4) \\ &\quad + \log \Gamma(168) - \log \Gamma(4). \end{aligned}$$

This gives

$$\boxed{\log p(D_Y \mid M) = -920.2423}.$$

For M_3 , `score(..., by.node=TRUE)` gives

Node	Contribution
X_1	-578.1982
X_2	-526.1777
X_3	-611.9504
Y	-920.2423

and

$$-578.1982 - 526.1777 - 611.9504 - 920.2423 = -2636.569,$$

which equals the total BDe log marginal likelihood for M_3 .

2.4.8 Marginal Likelihood versus Classification Accuracy

The model with the highest marginal likelihood does *not* have the highest hard-classification accuracy.

M_1 receives essentially all of the posterior model probability but has reported accuracy

$$0.5086,$$

while M_2 – M_5 each have accuracy

$$0.5131.$$

There is no contradiction because the two criteria answer different questions.

The marginal likelihood evaluates the model’s ability to describe the entire joint distribution,

$$p(Y, X_1, X_2, X_3),$$

whereas classification accuracy evaluates only whether

$$\arg \max_y P(Y = y \mid X_1, X_2, X_3, D)$$

matches the observed provider.

Thus, M_1 captures the dependence structure among the variables much better, but those additional dependencies rarely change which provider has the largest posterior probability.

3 Part C: Interpretation

The clearest individual attribute associated with provider preference is reputation satisfaction, X_1 . Among customers who prefer B, 65.0% report high reputation satisfaction, compared with 39.0% of those who prefer A and only 31.7% of those who prefer C. In contrast, price satisfaction varies relatively little across providers: 41.1% for A, 36.3% for B, and 36.6% for C. Service-variety satisfaction shows somewhat more differentiation than price, but much less than reputation.

The Bayesian-network model comparison also indicates that the attributes should not be treated as independent once provider preference is known. The fully dependent model M_1 receives essentially all of the posterior model probability. Thus, reputation, price, and service variety appear to contain important joint association patterns that are lost under the naive Bayes assumption.

For provider A, the results suggest that reputation may be a more important area for improvement than price. Customers preferring A have substantially lower high-reputation satisfaction than customers preferring B, while A’s service-variety satisfaction is relatively competitive. Increasing perceived reputation could therefore help A differentiate itself more strongly.

Provider C appears to face a more substantial challenge. It has the lowest high-reputation satisfaction rate of the three providers and also the lowest high service-variety satisfaction rate. Price does not appear to distinguish C strongly from the other providers. Therefore, the strongest opportunities suggested by the data for C are associated with reputation and, secondarily, service variety rather than price.

These results should be interpreted as associations rather than causal effects. The survey and Bayesian-network analysis identify attributes that are strongly related to provider preference, but they do not by themselves establish that changing an attribute would causally change customers’ provider choices.

Overall, reputation is the strongest individual differentiator, while the attributes also exhibit substantial joint dependence.